



Statistical Methods in Physics

FK7061 - Fall 2025

Lecturers: Jens Jasche, Arthur Loureiro // T.A.: Antoine Gilles Lordet

FK7061 Evaluation Quiz

**** This quiz will not count towards your grade. However, we ask that you take it seriously, as it will be used to help improve this course (and the pre-requisite courses) for future course offerings**

1. What is the difference between a Poisson Distribution and a Normal Distribution? Please provide an example scenario where a set of data would be reasonably fit by both distributions.
2. How is the Bayesian concept of probability different from a frequentist definition?
3. Can you state the three Kolmogorov axioms of probability, e.g., what properties do probabilities possess?
4. Can you write Bayes' law and comment on the prior probability used in Bayesian statistics?

Survey questions:

- Have you taken a statistics class (at the undergraduate or graduate level) before this course? If so, please specify the university and level that the class was taken at?
- What degree/specification are you pursuing at Stockholm University? Are you enrolled here, or as an ERASMUS student from another university?

Work Plan

The Frequentist Portion

Lecture 1: Introduction

Lecture 2: Probability Distributions

Lecture 3: Propagation of Errors, Covariances

Lecture 4: The Likelihood – Part 1

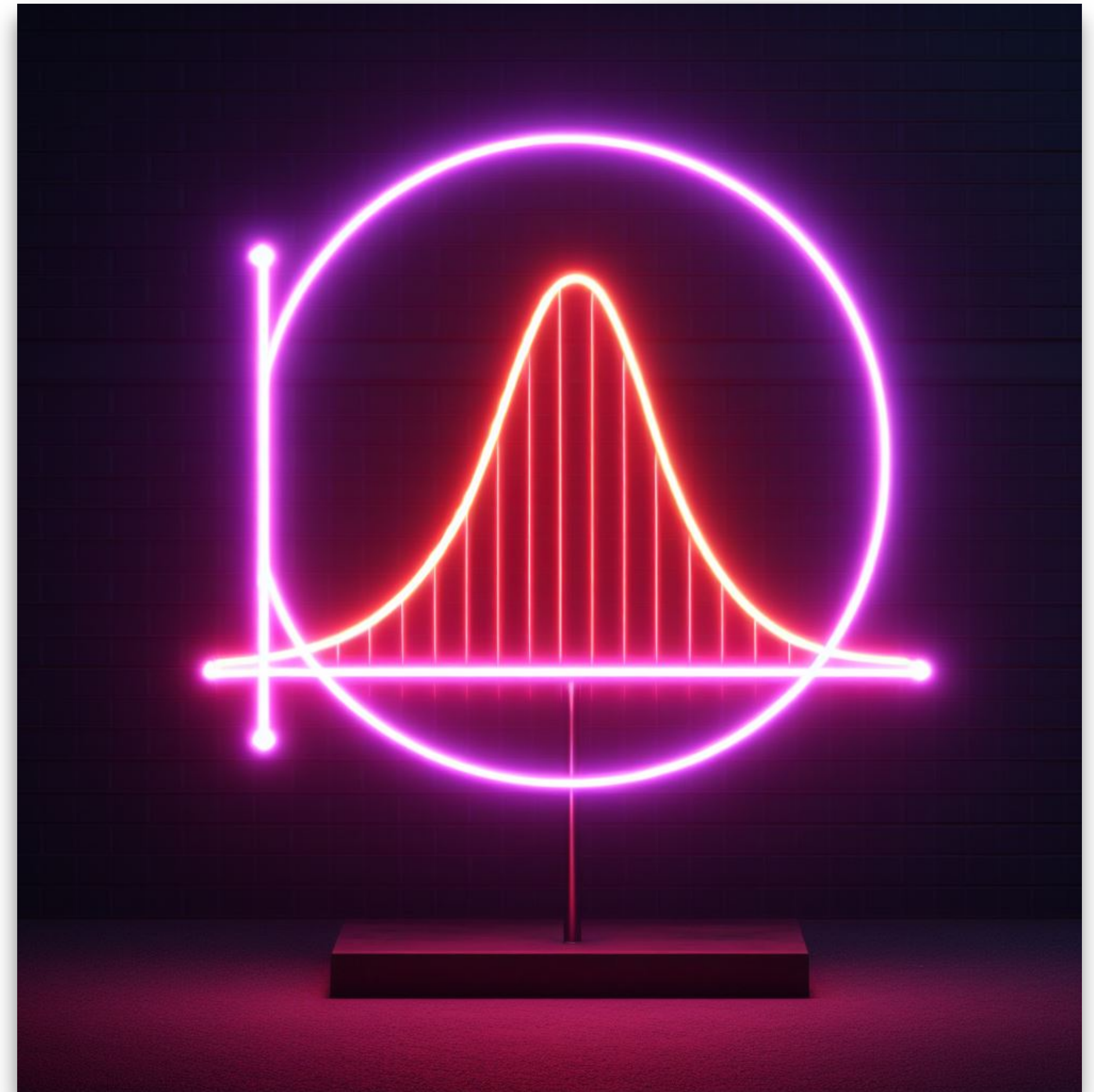
Lecture 5: The Likelihood – Part 2

Lecture 6: Method of Least Squares

Lecture 7: Confidence Intervals – Part 1

Lecture 8: Confidence Intervals – Part 2

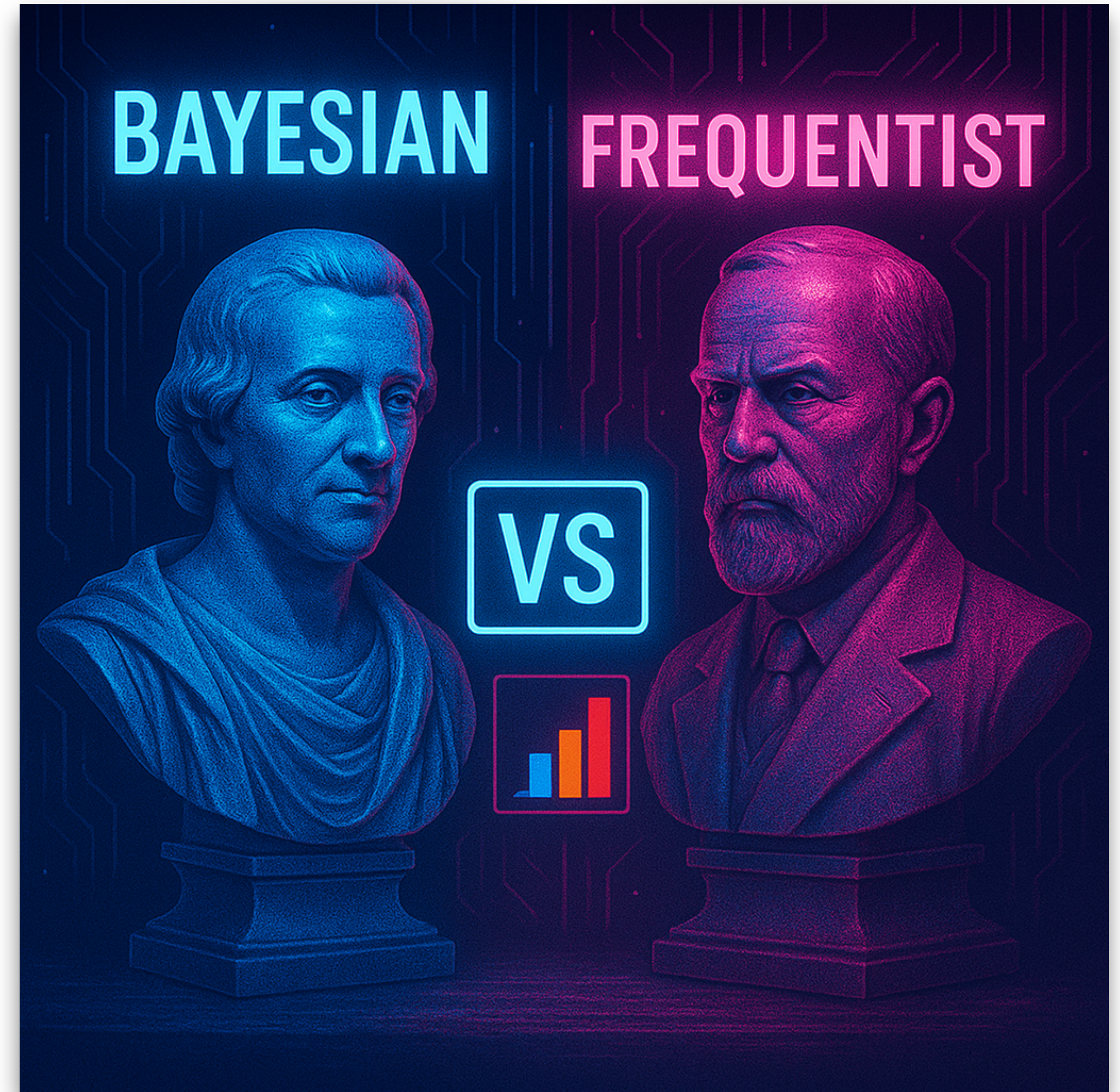
Lecture 9: Hypothesis Testing



Lecture 1

Outline

1. What is probability?
 - Popular definitions
 - Kolmogorov's Axioms
2. Random Variables
3. Probability Density Functions and Cumulative Distribution Functions
4. Mean and Standard Deviations
5. Interpretations of Probability
 - Frequentist
 - Bayesian



**What is your definition of
probability?**



Kolmogorov's Axioms

Kolmogorov's Axioms

Probability Calculus

Ω

Axiom #1 (First axiom of probability calculus): *Given that Ω is the universe of all possible outcomes of an experiment or the sample space. The joint probability of all true events in Ω must be maximal:*

$$\sum_{\forall E \in \Omega} P(E) = P(\Omega) = 1$$

Kolmogorov's Axioms

Probability Calculus

Ω

"Something has to happen"

Axiom #1 (First axiom of probability calculus): *Given that Ω is the universe of all possible outcomes of an experiment or the sample space. The joint probability of all true events in Ω must be maximal:*

$$\sum_{\forall E \in \Omega} P(E) = P(\Omega) = 1$$

Kolmogorov's Axioms

Probability Calculus

Ω

Axiom #2 (Second axiom of probability calculus):

$$\forall A \subseteq \Omega, P(A) \geq 0$$

Kolmogorov's Axioms

Probability Calculus

Ω

“The probability of an event
can *never* be negative”

Axiom #2 (Second axiom of probability calculus):

$$\forall A \subseteq \Omega, P(A) \geq 0$$

Kolmogorov's Axioms

Probability Calculus

Ω

Axiom #3 (Third axiom of probability calculus):

$$\forall A, B \subseteq \Omega, \text{ if } A \cap B = \emptyset, \text{ then } P(A \cup B) = P(A) + P(B)$$

Kolmogorov's Axioms

Probability Calculus

Ω

“Probabilities of different
(disjoint) events add”

Axiom #3 (Third axiom of probability calculus):

$$\forall A, B \subseteq \Omega, \text{ if } A \cap B = \emptyset, \text{ then } P(A \cup B) = P(A) + P(B)$$

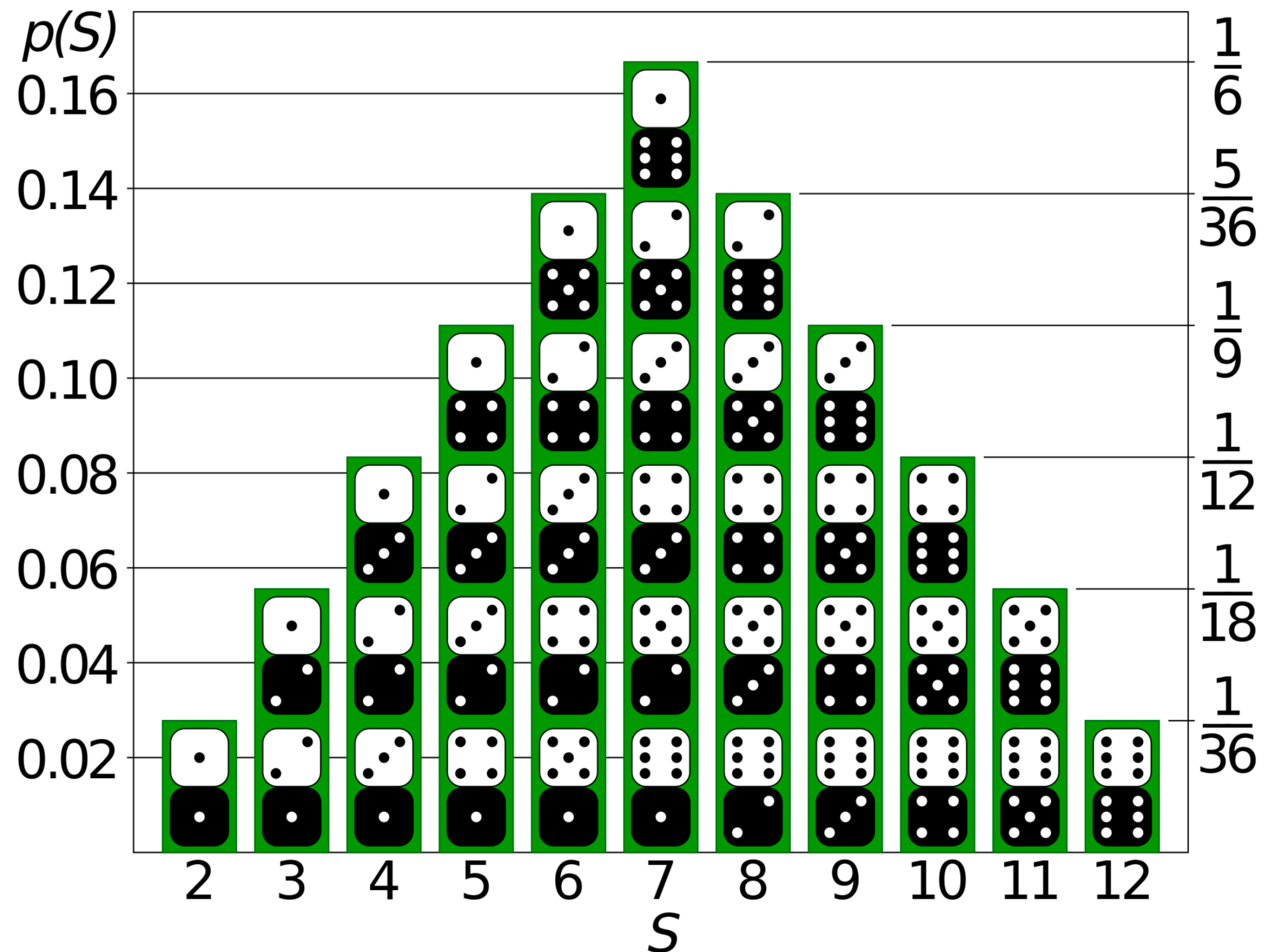
Random Variables

A measurable function defined on a probability space (which obeys the Kolmogorov axioms)



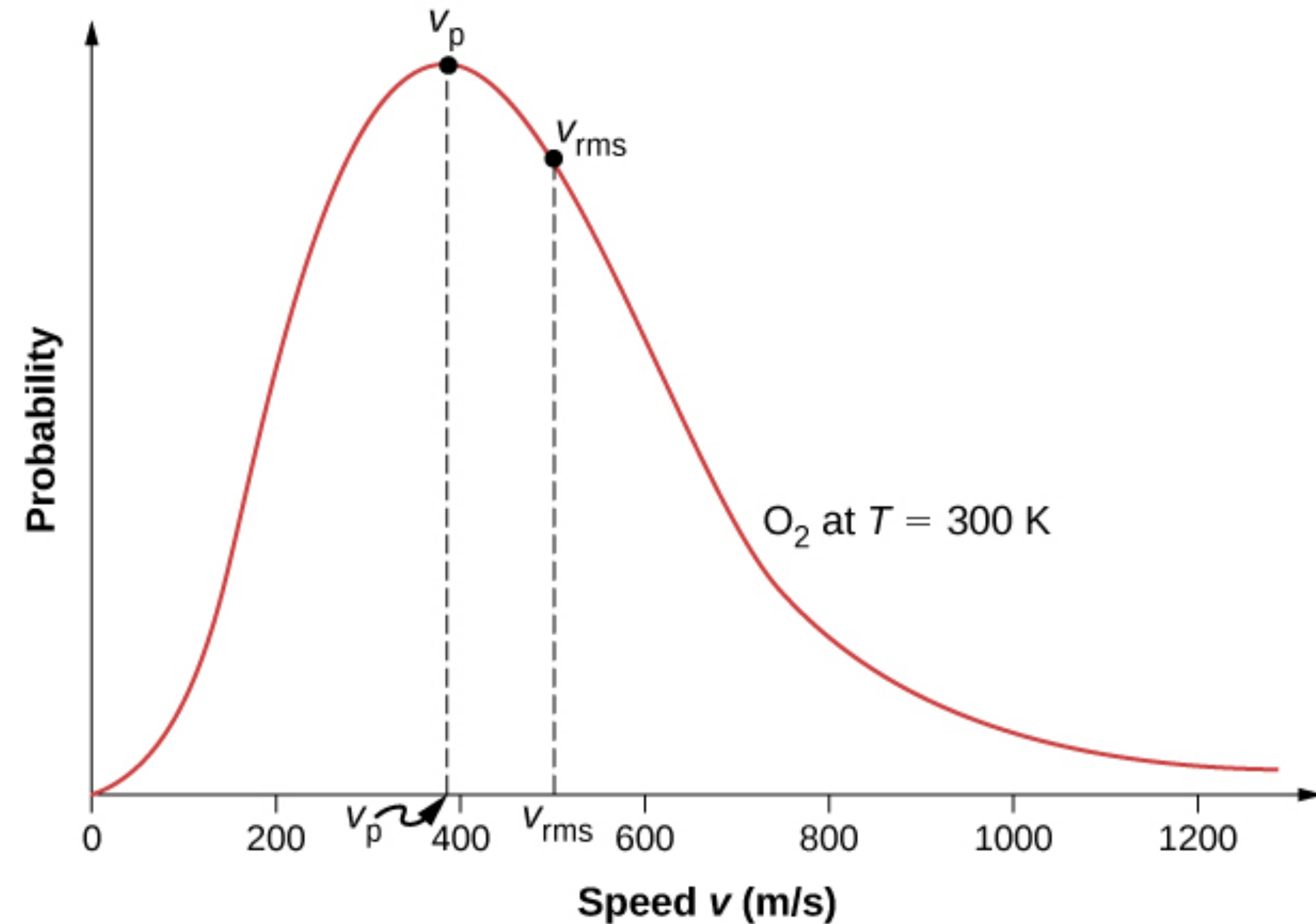
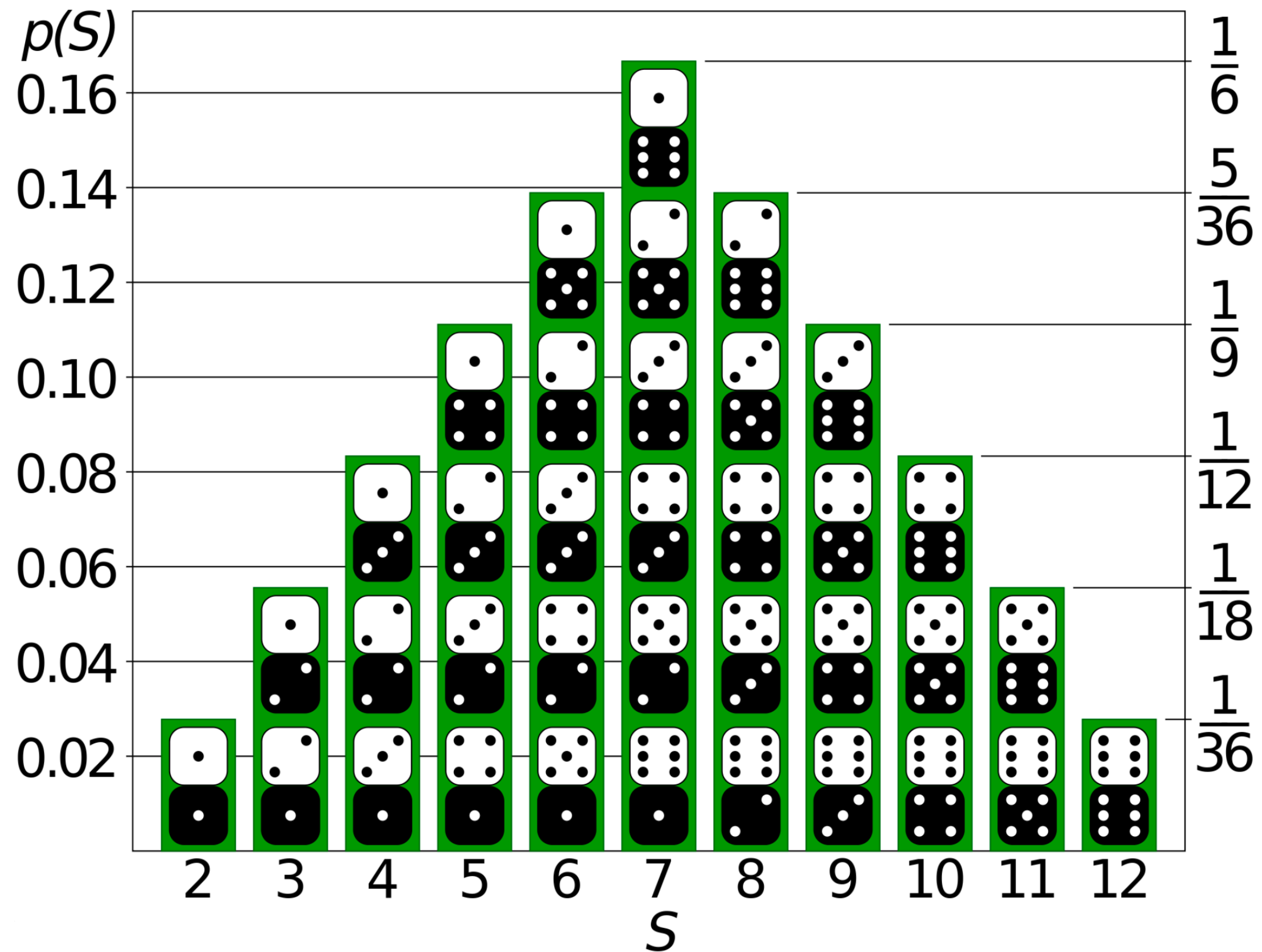
Random Variables

A measurable function defined on a probability space (which obeys the Kolmogorov axioms)



Random Variables

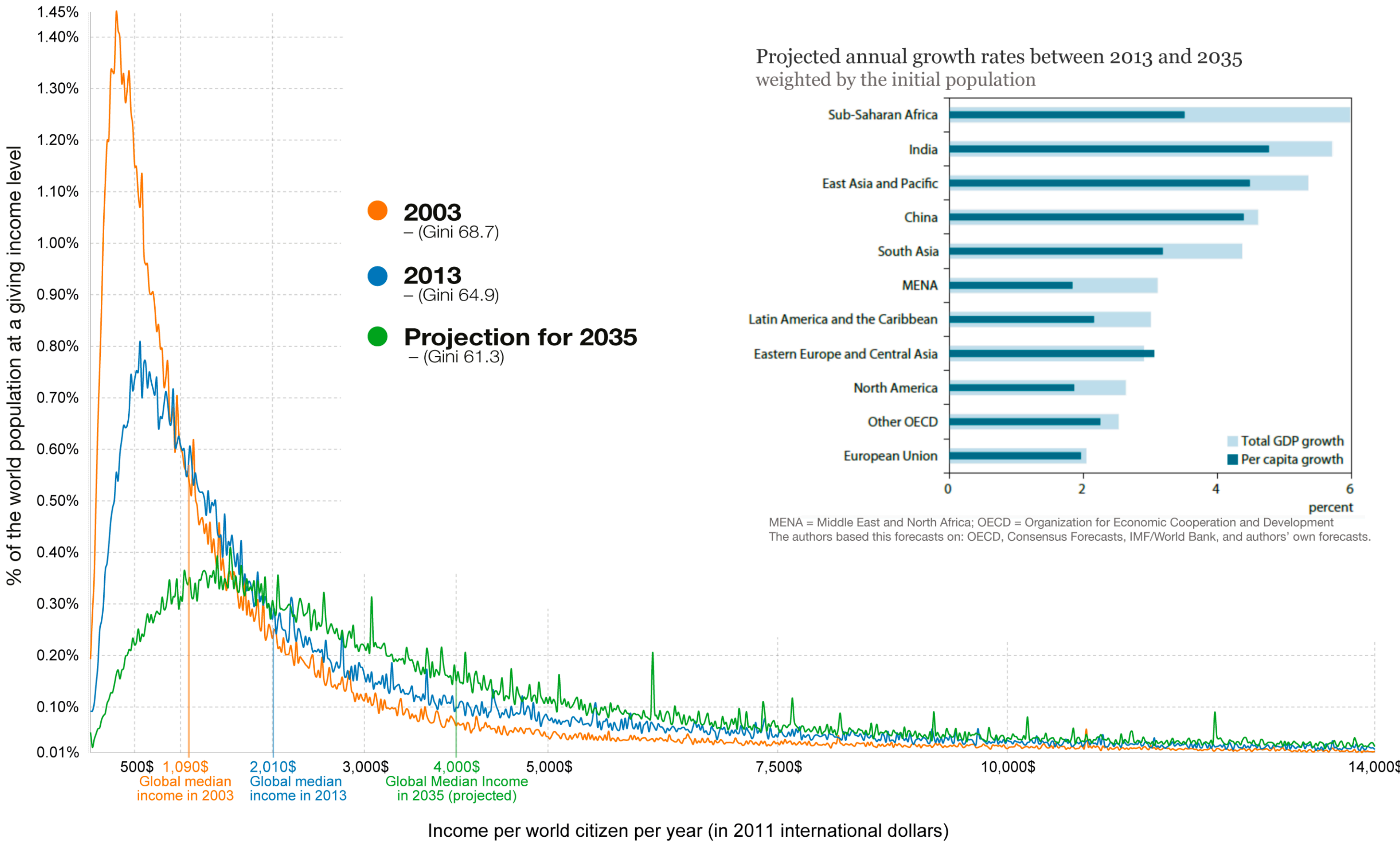
Random variables can be discrete or continuous



Random Variables

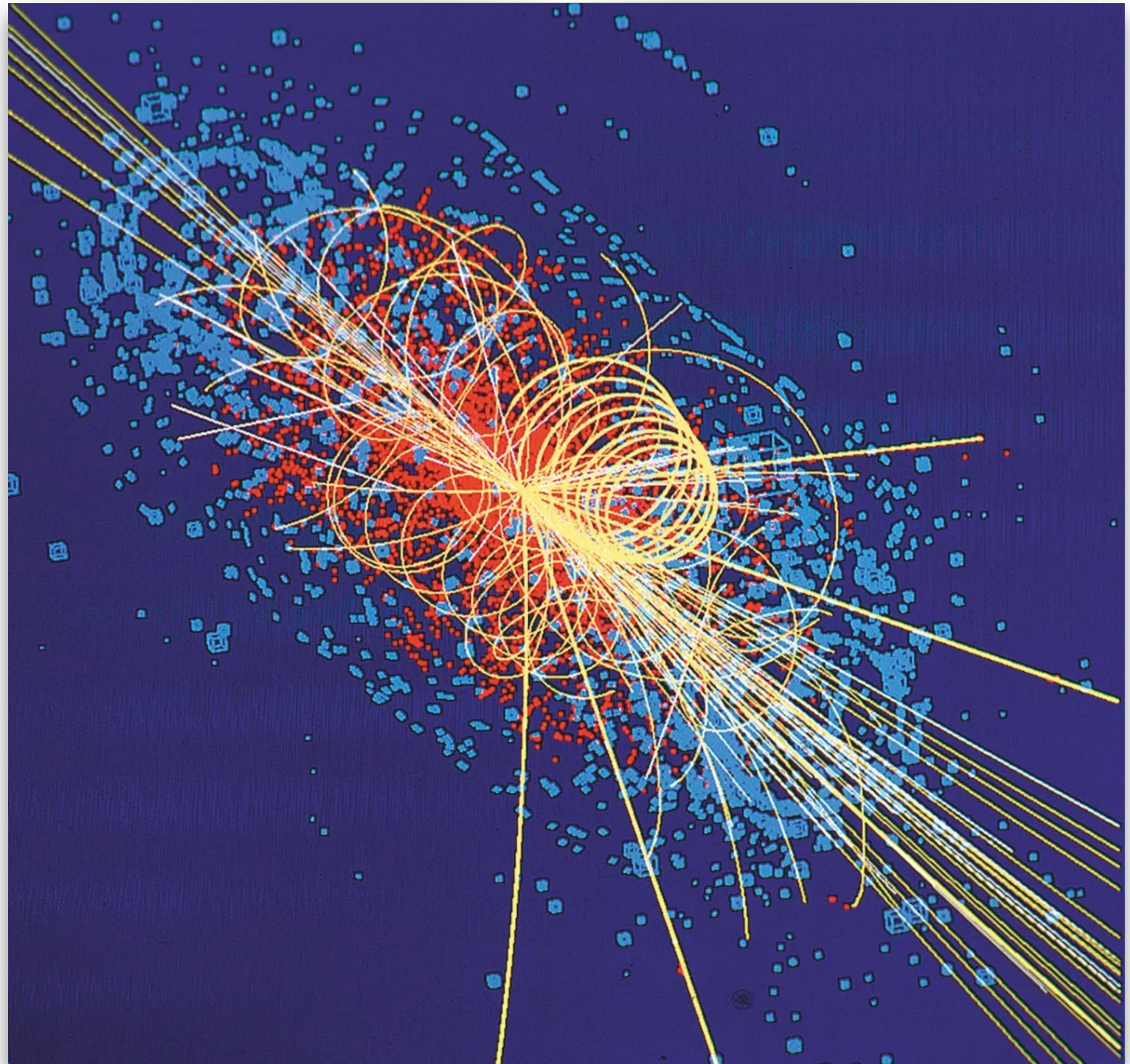
Or in between...

The global income distribution in 2003, 2013, and the projection for 2035
Incomes are adjusted for price changes over time and for price differences between countries (purchasing power parity (PPP) adjustment).



Random Variables

Random variables can be multidimensional (Lecture 3)



Some Nomenclature and Notation

- $\Omega = \{\text{the set of all possible events}\}$
- $A \cup B = \{\text{all events that are in either } A \text{ or } B\} - \textit{union}$
- $A \cap B = \{\text{all events that are in } A \text{ and } B\} - \textit{intersection}$
- $A \subset B = \{\text{all events in } A \text{ are also in } B; A \text{ is a subset of } B\} - \textit{contained in}$
- A^C or $\bar{A} = \{\text{all events that are not in } A\} - \textit{complement}$

Probability Spaces

- $P(\Omega) = 1$

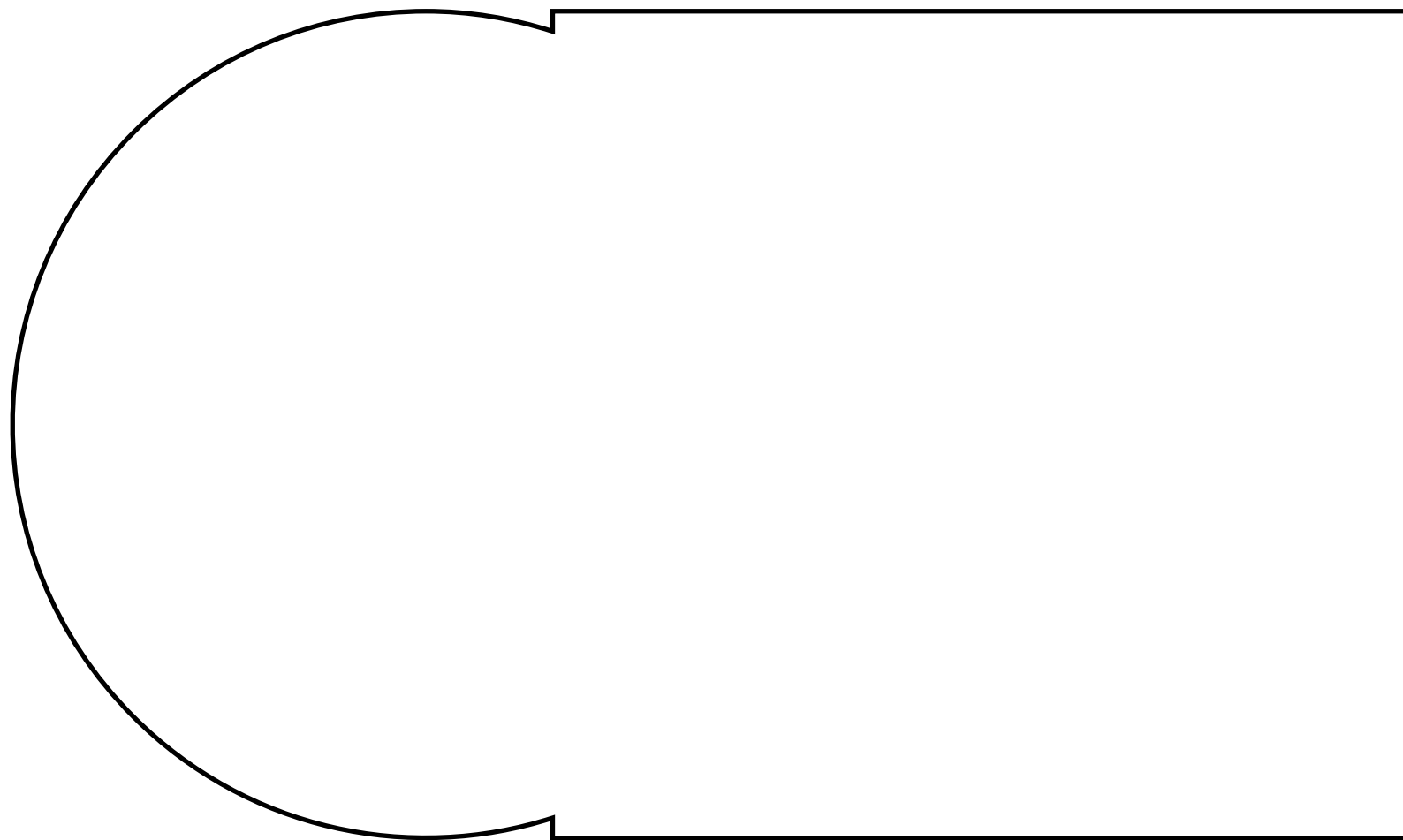
Ω

Probability Spaces

Ω

Probability Spaces

- $P(A \cup B)$

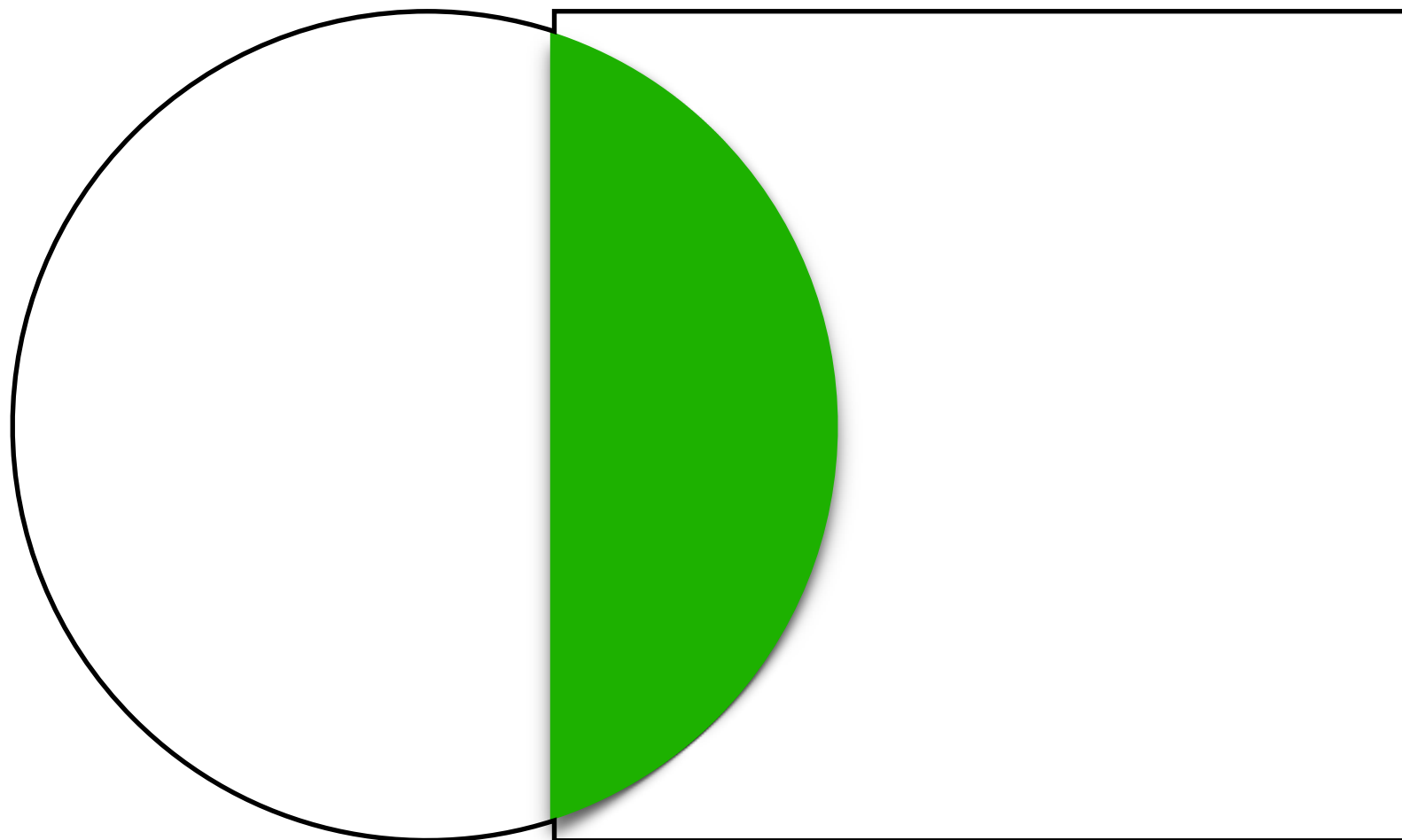


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Probability Spaces

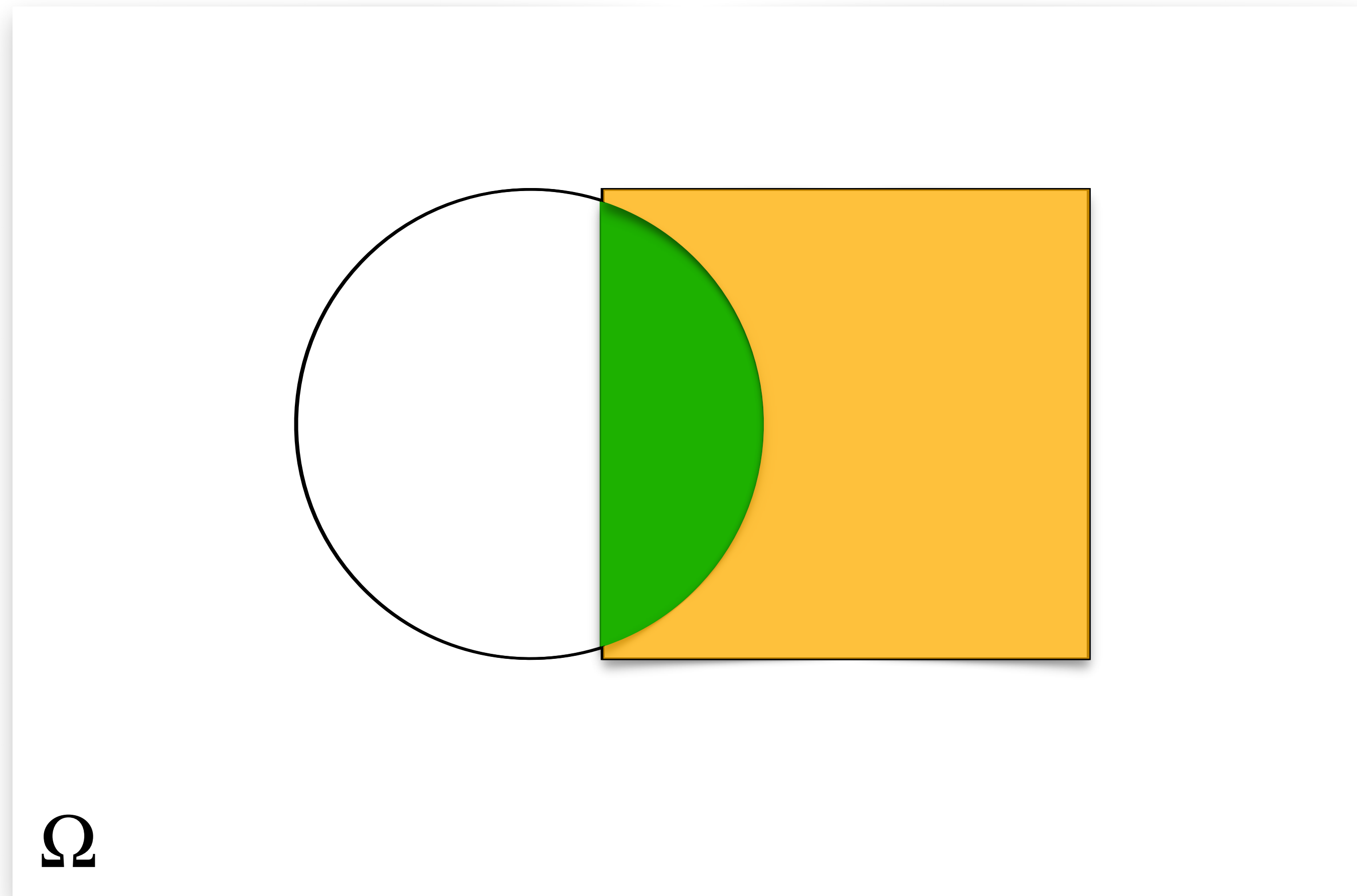
- $P(A \cup B)$

- $P(A \cap B)$



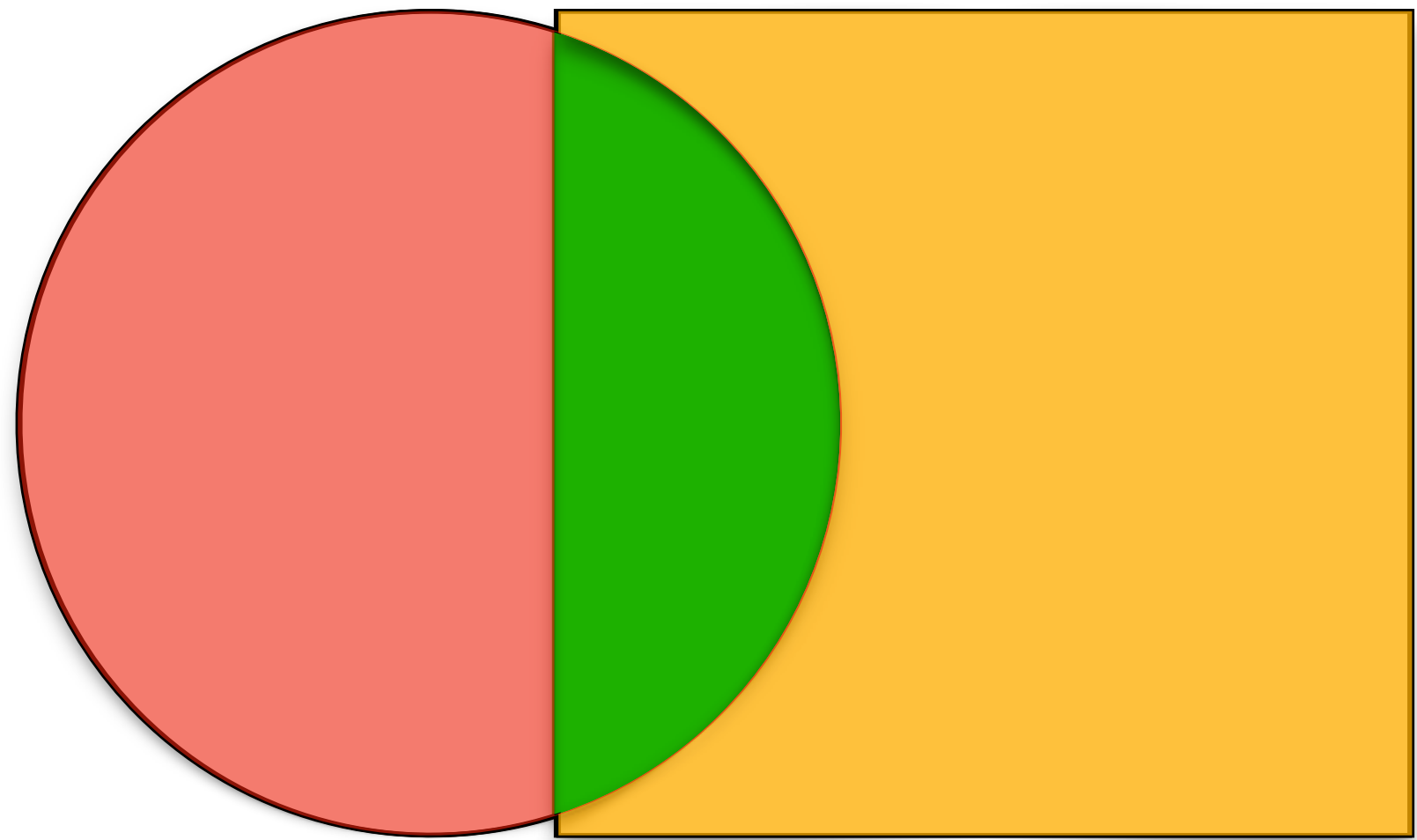
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Probability Spaces



- $P(A \cup B)$
- $P(A \cap B)$
- $P(B \cap \bar{A})$

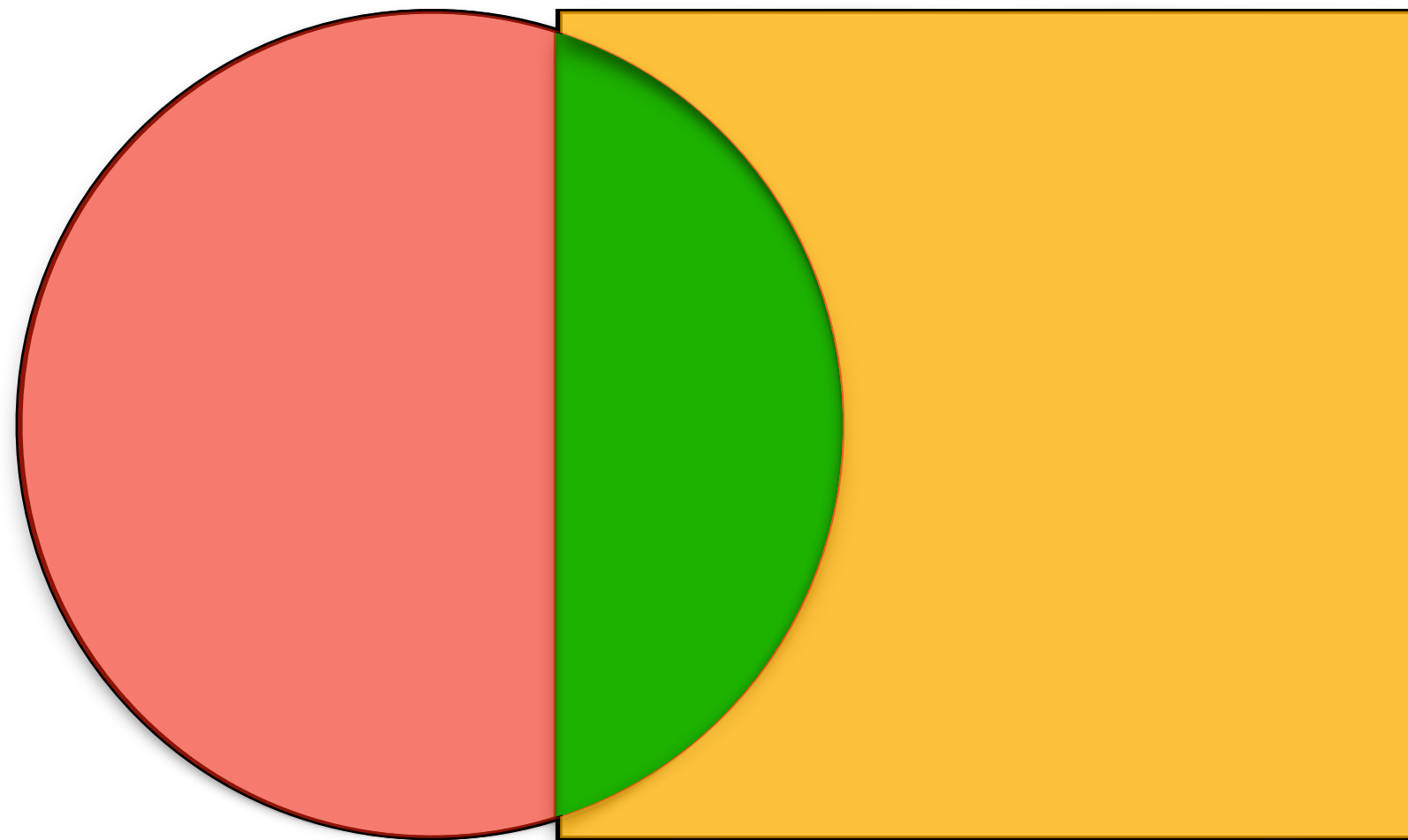
Probability Spaces



Ω

- $P(A \cup B)$
- $P(A \cap B)$
- $P(B \cap \bar{A})$
- $P(A \cap \bar{B})$

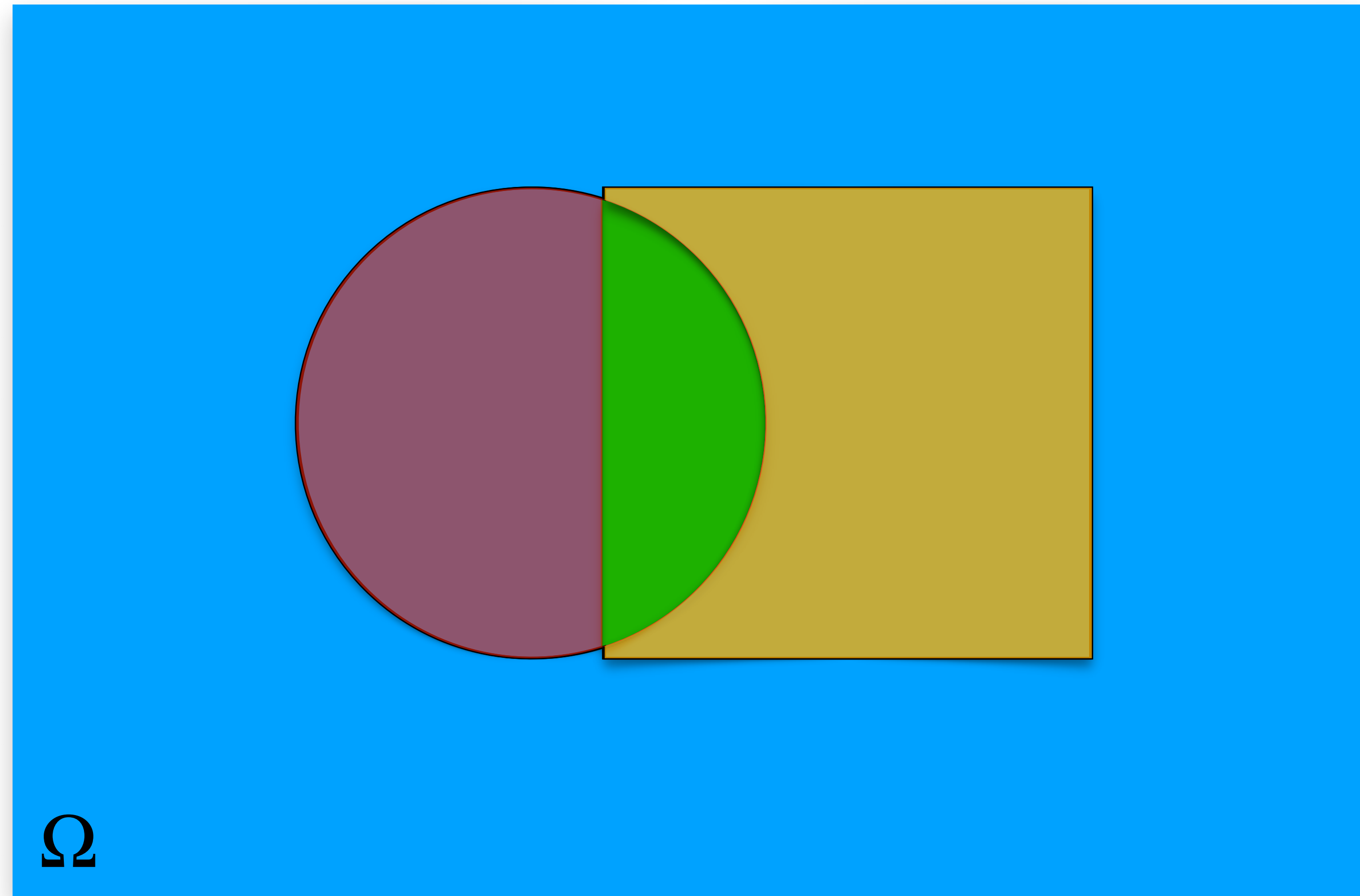
Probability Spaces



Ω

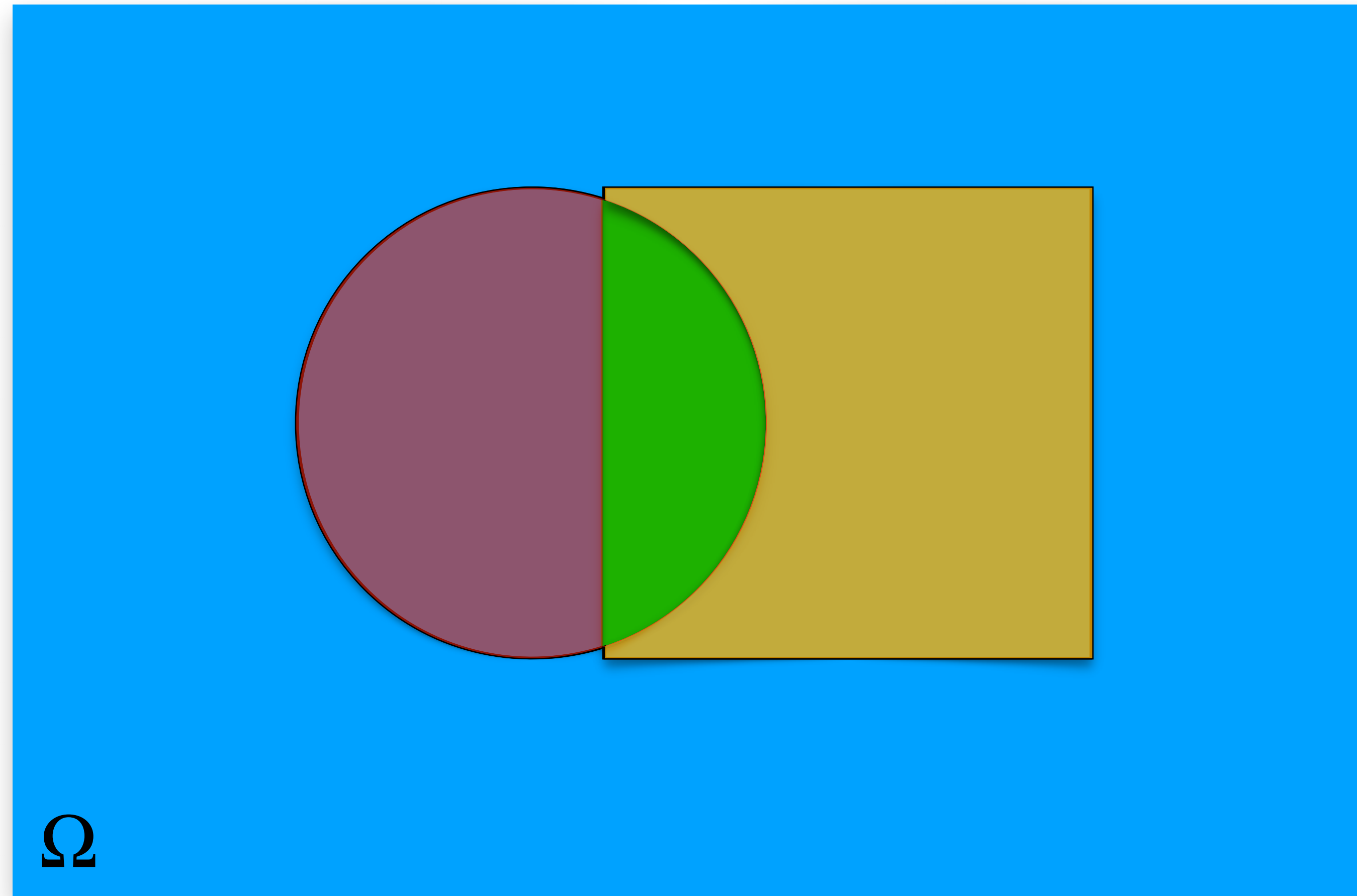
- $P(A \cup B)$
- $P(A \cap B)$
- $P(B \cap \bar{A})$
- $P(A \cap \bar{B})$
- $P(\bar{A} \cap \bar{B})$

Probability Spaces



- $P(A \cup B)$
- $P(A \cap B)$
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- $P(A \cap \bar{B})$
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Probability Spaces



- $P(A \cup B)$
- $P(A \cap B)$
- $P(B \cap \bar{A})$
- $P(A \cap \bar{B})$
- $P(\bar{A} \cap \bar{B})$

$$P(\Omega) = 1 = P(A \cap B) + P(A \cap \bar{B}) + P(B \cap \bar{A}) + P(\bar{A} \cap \bar{B})$$

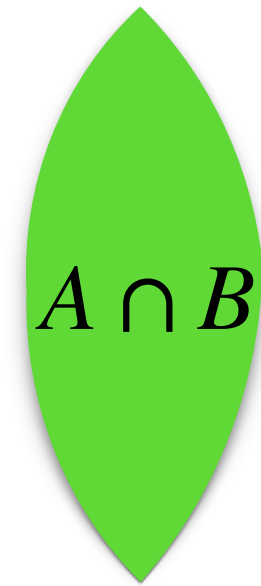
Probability Spaces

Ω

If $A \subset B$ then $P(A \cap B) = P(A)$

Probability Spaces

Conditional Probability



Ω

$$P(A | B) = \frac{A \cap B}{B}$$

$P(A | B)$ = {“conditional probability” of an event being in A, given that the event is in B}

Dependent and Independent Events

Two variables are independent if: $P(A \cap B) = P(A)P(B)$

(Assume the probability is equally distributed through the space)



$$P(A \cap B) = P(A) \neq P(A)P(B)$$

Dependent and Independent Events

Two variables are independent if: $P(A \cap B) = P(A)P(B)$

(Assume the probability is equally distributed through the space)



$$P(A \cap B) = 1/3 \times 1/3 = 1/9 = P(A)(B)$$

Dependent and Independent Events

Two variables are independent if: $P(A \cap B) = P(A)P(B)$

(Assume the probability is equally distributed through the space)



$$P(A \cap B) = 0 \neq P(A)P(B)$$

Dependent and Independent Events

Two variables are independent if: $P(A \cap B) = P(A)P(B)$

(Assume the probability is equally distributed through the space)



$$P(A)P(B) = P(A) \times 0.25 \neq P(A \cap B)$$

Let's do some simple proofs

- Exercise I: Prove that if $A \subset B$, then $P(A) \leq P(B)$

 See Lecture 1 Notes , pg 1

- Exercise II: Prove that $P(A \cup B) = P(A) + P(B) - P(A \cap B)$

 See Lecture 1 Notes , pg 2

Break Time :)

Dropping the Bayes



Bayes' Theorem

The diagram illustrates Bayes' Theorem with the following components and callouts:

- Posterior Distribution:** $P(A | B)$ (enclosed in an orange box)
- Likelihood:** $P(B | A)$ (enclosed in a green box, with a callout: "Likelihood of the data (observation) given A (a parameter we wish to infer). It describes a generative model by which the observation is generated given A.")
- Prior:** $P(A)$ (enclosed in a red box, with a callout: "Background information we have about A: either reflecting our belief about the parameter or the known prevalence (base rates)")
- Evidence:** $P(B)$ (enclosed in a blue box, with a callout: "A normalisation factor which can be seen as 'the probability of the observation'. This is usually ignored but relevant for model comparison!")

$$P(A | B) = \frac{P(B | A)P(A)}{P(B)}$$

Bayes' Theorem

$$\underbrace{P(A | B)}_{\text{Posterior Distribution}} = \frac{\overbrace{P(B | A)}^{\text{Likelihood}} \overbrace{P(A)}^{\text{Prior}}}{\underbrace{P(B)}_{\text{Evidence}}}$$

Bayes' Law (or Bayes' Theorem) describes an update of the plausibility $P(A)$ to a new posterior degree of plausibility $P(A | B)$ given some new observation B .

Bayes' Theorem

$$\underbrace{P(A | B)}_{\text{Posterior Distribution}} = \frac{\overbrace{P(B | A)}^{\text{Likelihood}} \overbrace{P(A)}^{\text{Prior}}}{\underbrace{P(B)}_{\text{Evidence}}}$$

Note that Bayes' theorem is independent of Bayesian Statistics!

Probability Density Function

Let's define $f(x) \equiv p(x)$ as the density of probabilities in some space.

Hence:

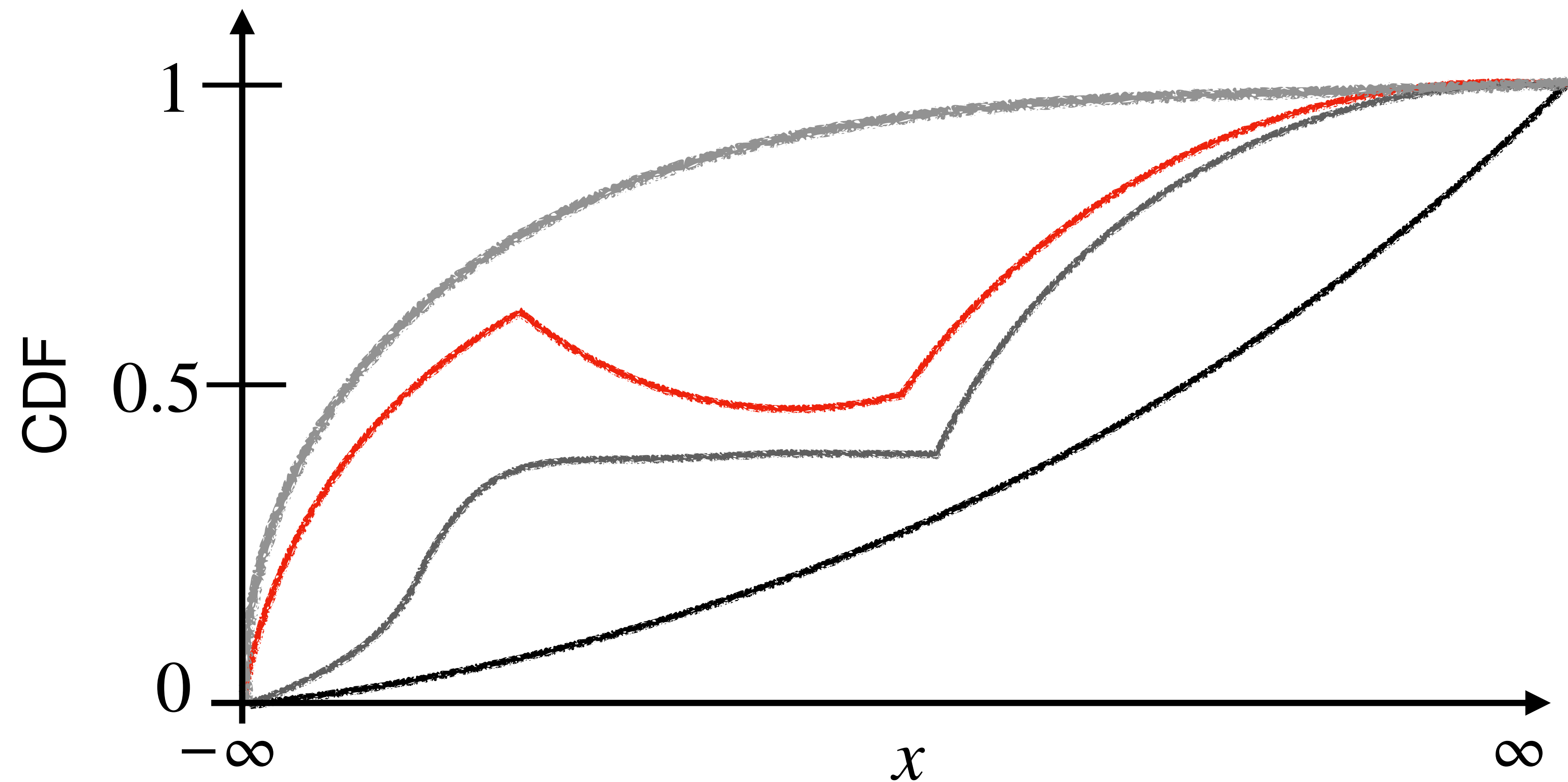
$$P(a < x < b) = \int_a^b f(x)dx$$

Following Kolmogorov's axioms:

$$P(-\infty < x < \infty) = \int_{-\infty}^{\infty} f(x)dx = 1$$

Cumulative Density Function

Let's define $F(x) = \int_{-\infty}^x f(x')dx' \equiv P(x' < x)$ as the cumulative density function



Redefining Standard Statistics

In terms of PDFs and CDFs

- **Median:** $y_{1/2} = F^{-1}(0.5)$
- **Mode:** $\operatorname{argmax}[f(x)]$
- **Mean (expectation value):** $\langle x \rangle = \int_{-\infty}^{\infty} x f(x) dx$

Expectation values and Variance

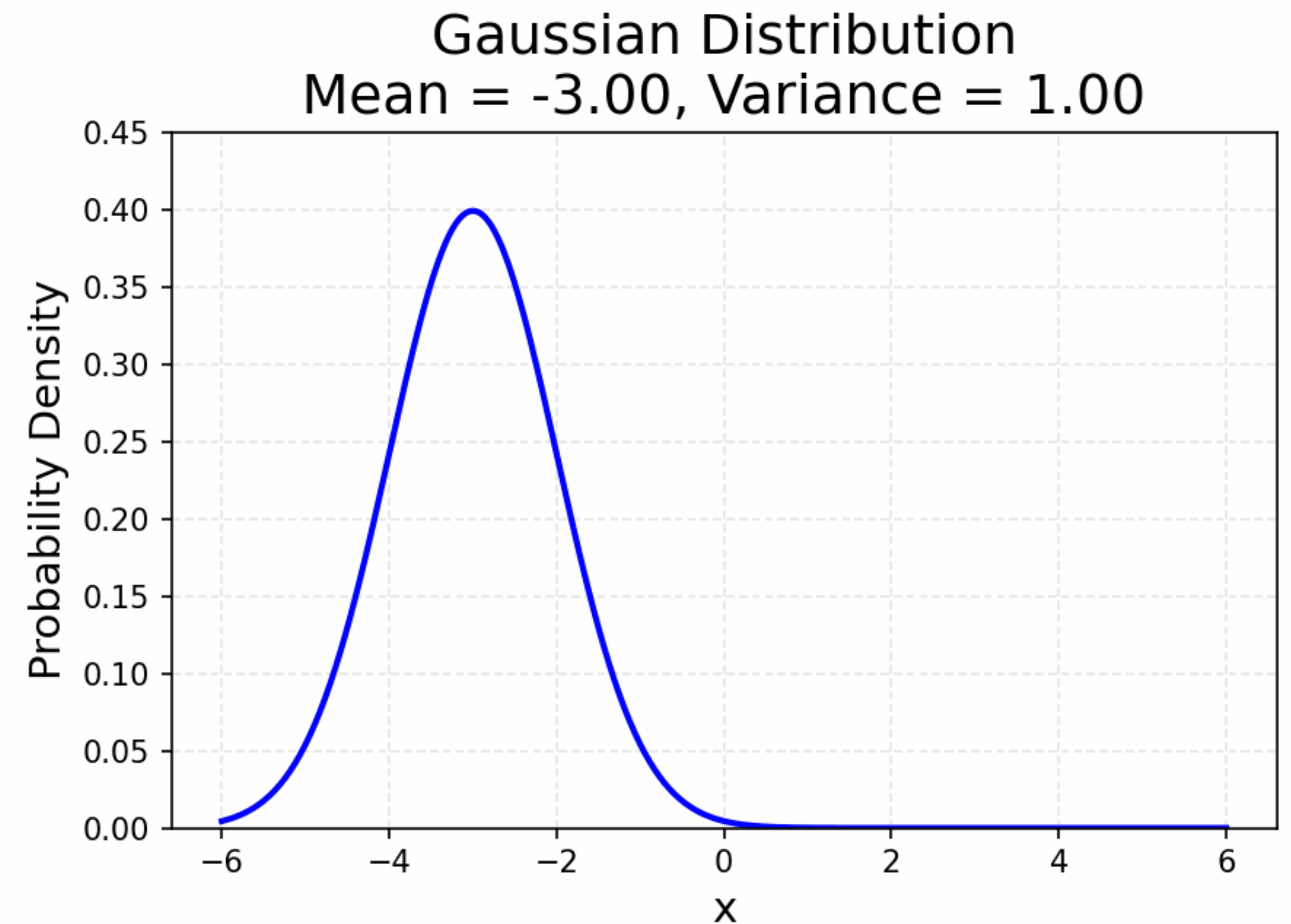
Definitions

- **Expectation:**

$$E[X] = \mu = \int_{x_{\min}}^{x_{\max}} x f(x) dx$$

- **Variance:**

$$V[X] = \sigma^2 = E[(X - \mu)^2] = \int_{x_{\min}}^{x_{\max}} (x - \mu)^2 f(x) dx$$



Expectation values and Variance

We can also calculate higher-order moments

- **Skewness:**

$$S[X] = E[(X - \mu)^3] = \int_{x_{\min}}^{x_{\max}} (x - \mu)^3 f(x) dx$$

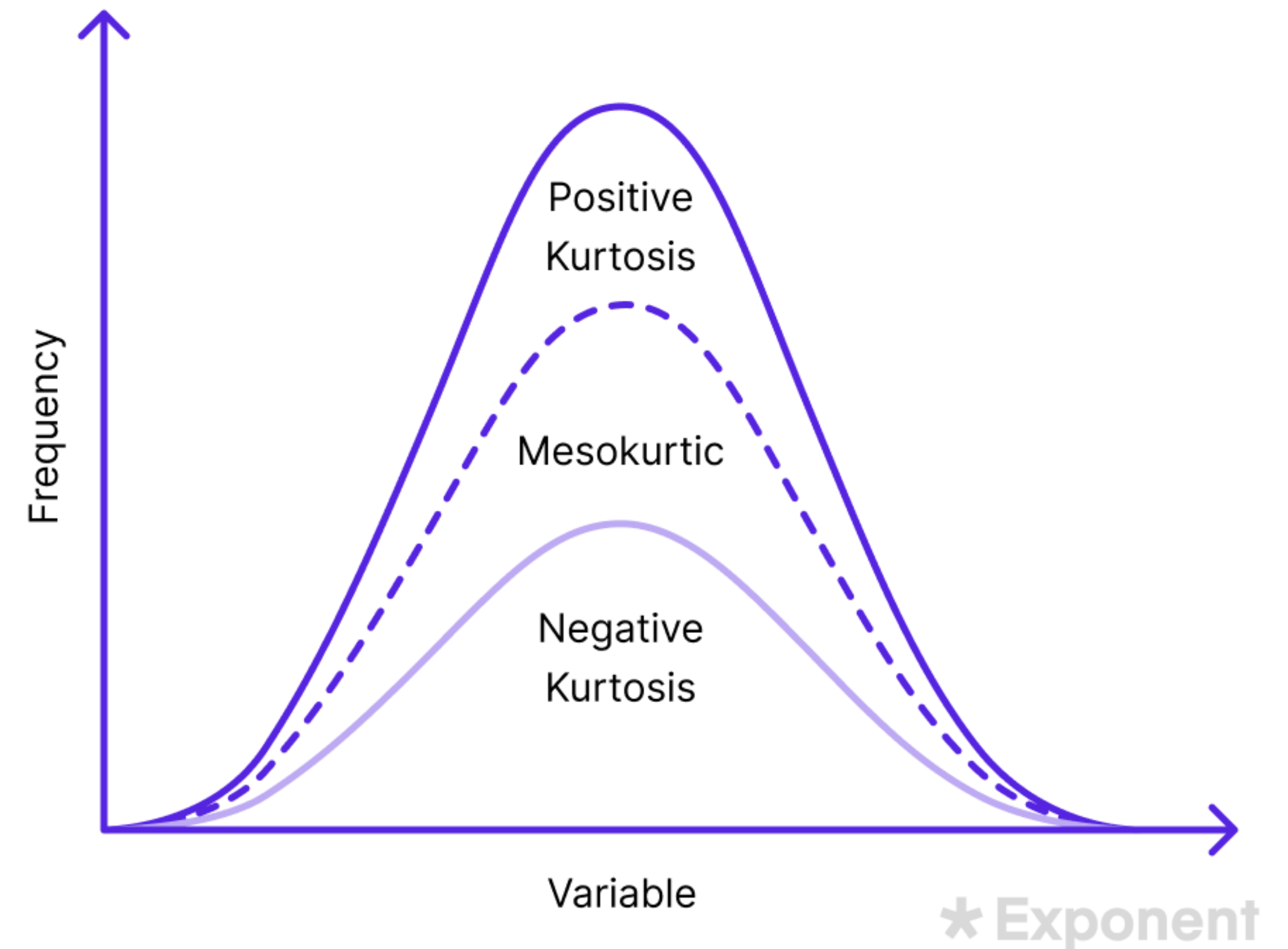
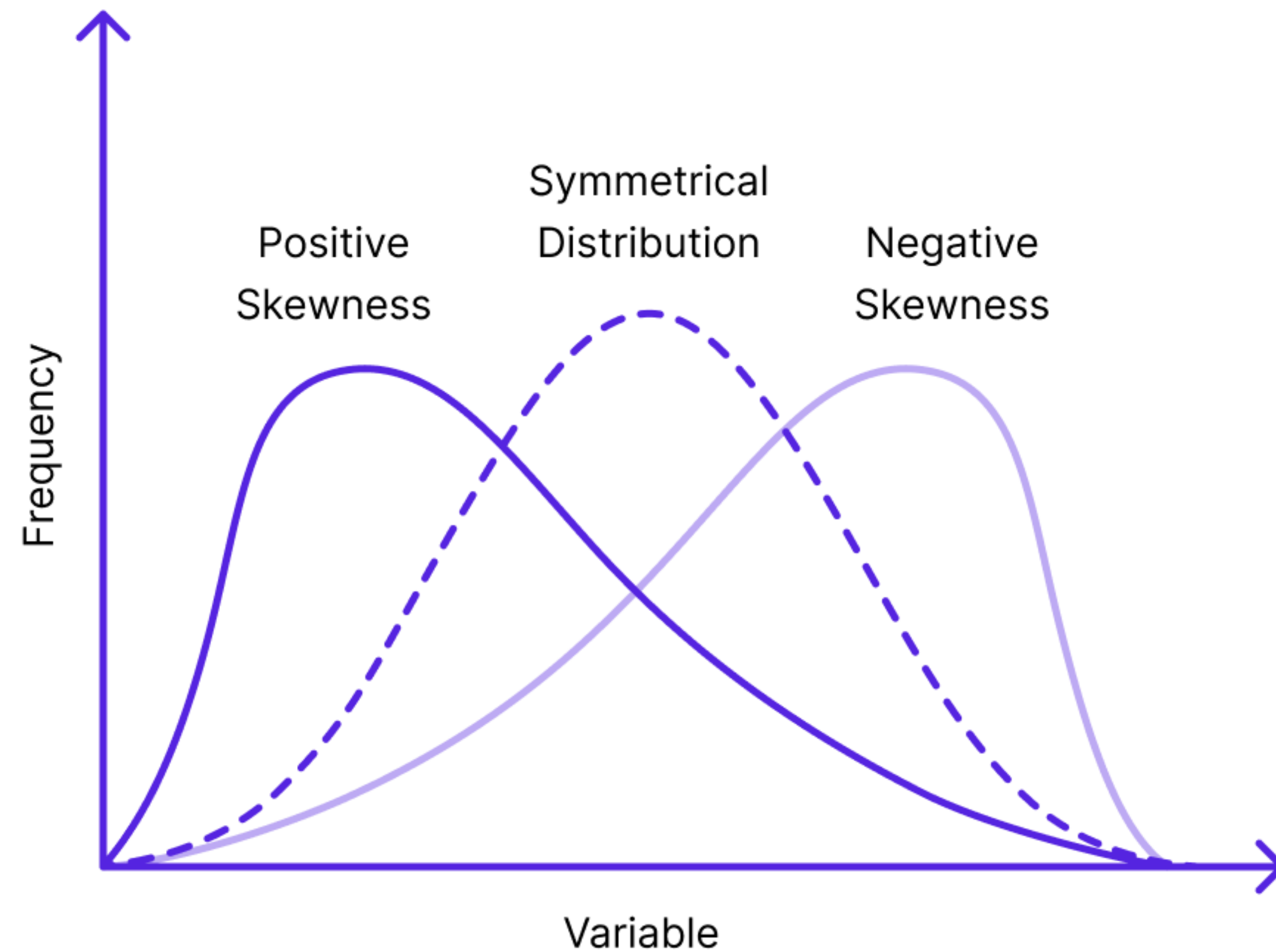
- **Kurtosis:**

$$K[X] = E[(X - \mu)^4] = \int_{x_{\min}}^{x_{\max}} (x - \mu)^4 f(x) dx$$

Expectation values and Variance

We can also calculate higher-order moments

Skewness & Kurtosis



Characteristic Functions

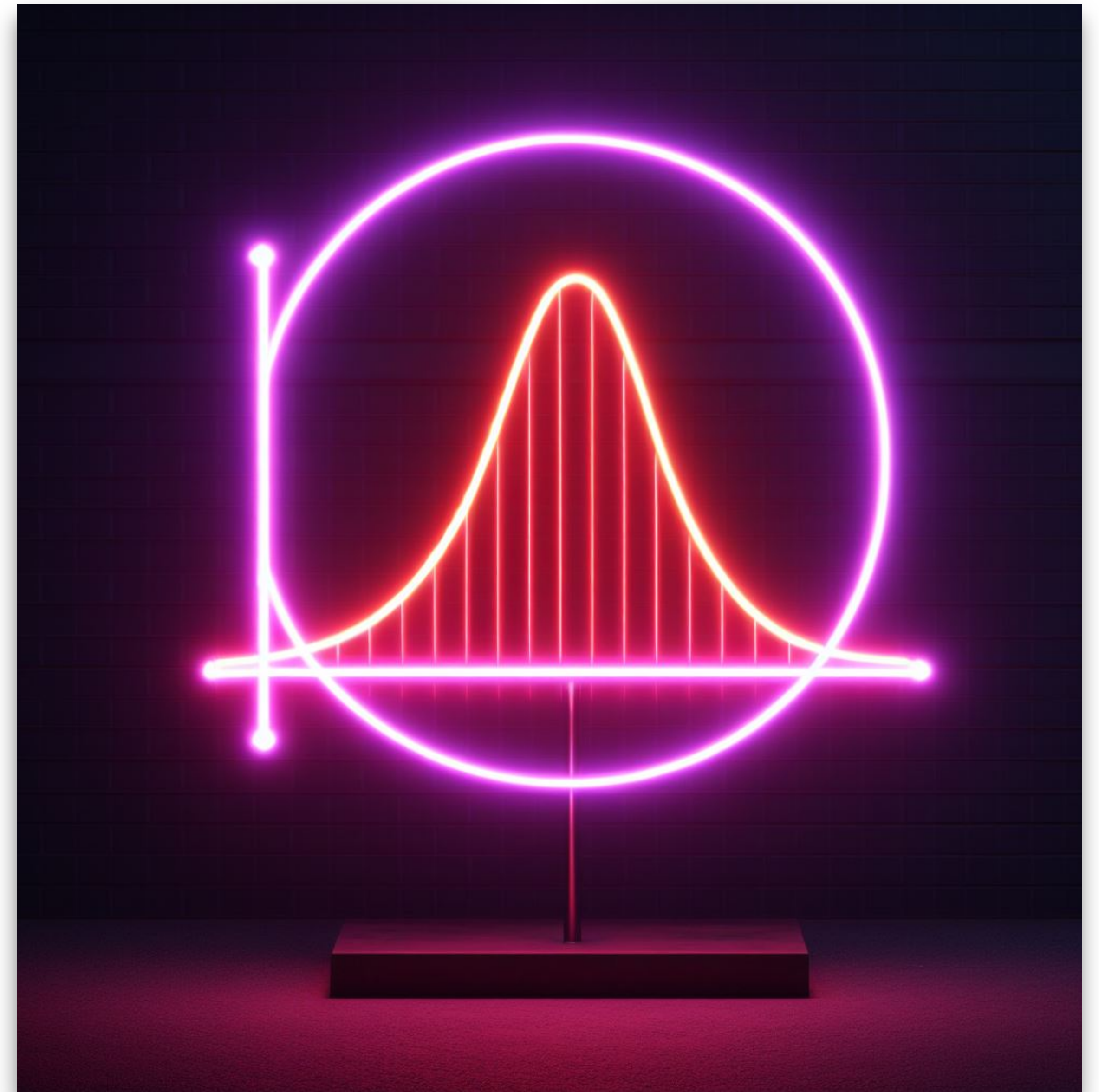
- Essentially the Fourier transform of the probability distribution function:

$$\phi_X(t) = \int_{-\infty}^{\infty} e^{itx} f(x) dx$$

- This can be used to describe and differentiate between different classes of probability density functions.

Expectation Values on Probability Space

- Earlier on, we showed that you can calculate different moments and characteristic functions on a probability space.
- Even more generically - You can calculate the expectation value of any function on a probability space



Expectation Values on Probability Space

Example: Let's assume that cars are speeding on a street, and the velocity that they are moving (over the speed limit) is given by:

$$f(x) = \frac{c}{x}, \quad 1 \leq x \leq 10$$

There are police officers that might pull these cars over, maybe multiple times! How likely this is depends on the cars speed, following a function:

$$g(x) = \frac{1}{200}x^2$$

How many times does the average car get pulled over?



Expectation Values on Probability Space

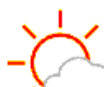
$$E[g(x)] = \int_{-\infty}^{\infty} g(x)f(x)dx$$

Popular Probability

How would you describe these probabilities to a friend?

- **Frequentist:** Our Model, repeated many times, will include the real value most* of the time.
- **Bayesian:** Expression of our ignorance, given our current Model and our prior knowledge.

10-DAY WEATHER FORECAST

TODAY 16/08	 19° 12°	Windy and not as warm  Night: Breezy early; cloudy
SUN 17/08	 20° 11°	Sunny breaks and breezy  Clear
MON 18/08	 23° 14°	Pleasant with sunny spells  Partly cloudy
TUE 19/08	 17° 9°	Cooler with broken intervals  Clear to partly cloudy
WED 20/08	 17° 9°	Broken intervals  Partly cloudy
THU 21/08	 18° 9°	Bright periods  Mainly clear
FRI 22/08	 17° 8°	Partly sunny  Broken intervals
SAT 23/08	 16° 9°	Broken cloud and sunny spells  Clear to partly cloudy
SUN 24/08	 17° 10°	Partly sunny  Clear
MON 25/08	 17° 10°	A little morning rain  Cloudy

☔ 10%

☔ 12%

☔ 4%

☔ 25%

☔ 16%

☔ 25%

☔ 25%

☔ 25%

☔ 7%

☔ 55%

Interpretations of Statistics

The Fundamental Problem:

We can measure $P(\text{Data} \mid \text{Model 1})$ and $P(\text{Data} \mid \text{Model 2})$ – but what we really want to know is $P(\text{Model 1} \mid \text{Data})$ and $P(\text{Model 2} \mid \text{Data})$

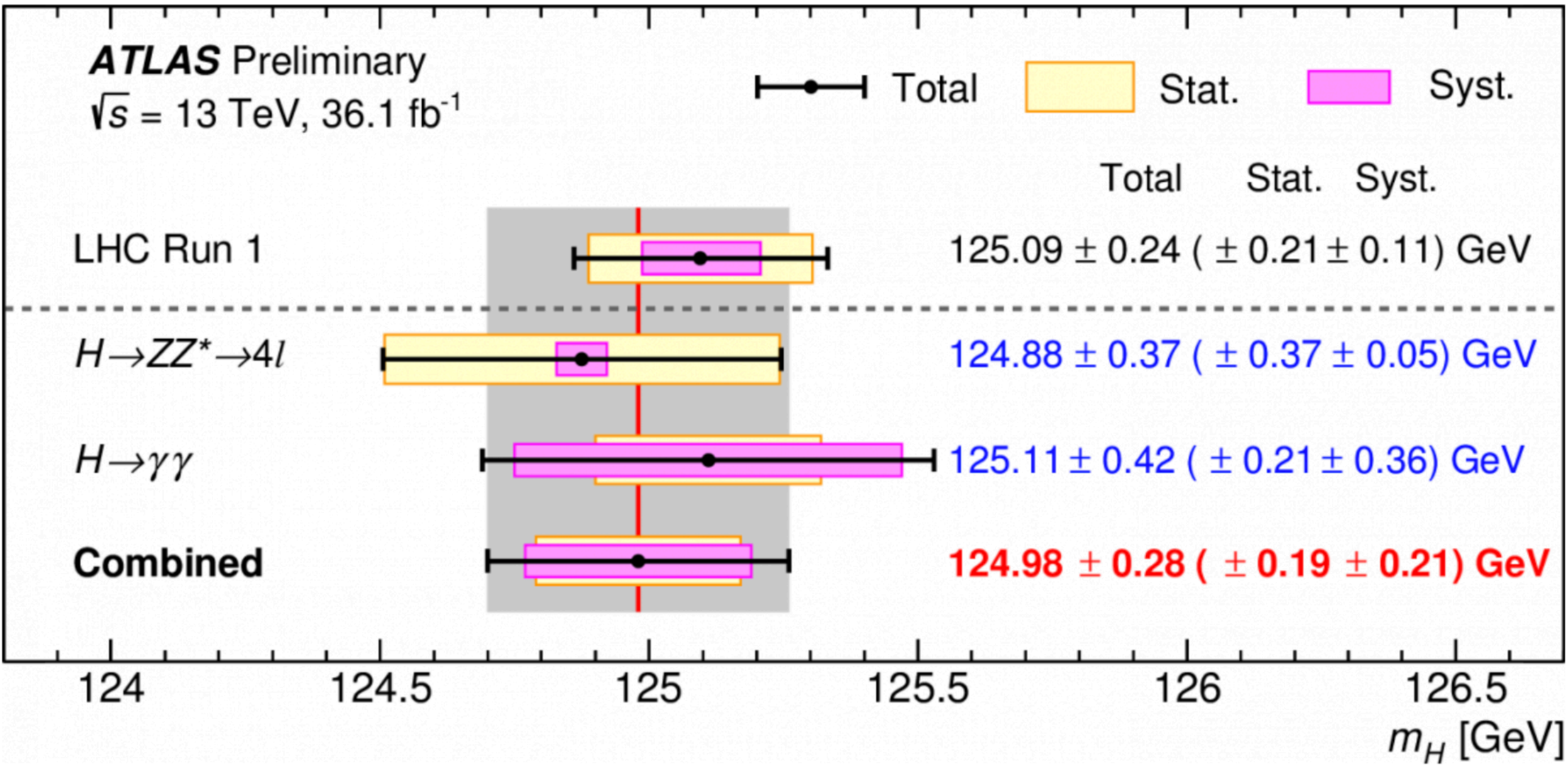
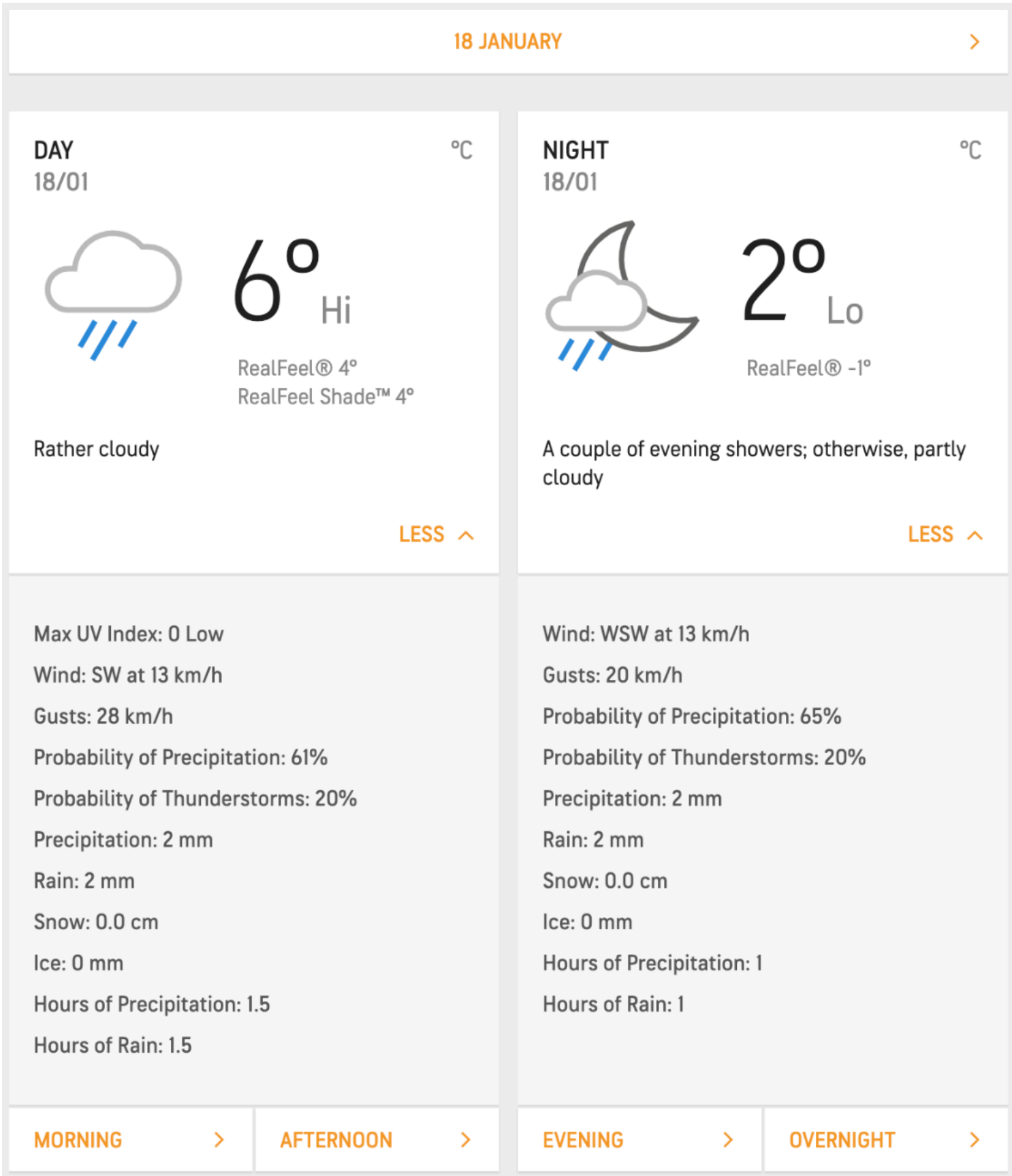
General Problem in Physics:

It is often straightforward to predict a measurement from physical theory — but we usually want to use data to determine if our theory is right.

Interpretations of Statistics

Popular Understanding (Wrong!)

“P(A) is the probability that A is true.”



Interpretations of Statistics

Frequentist Statistics

- $P(A)$ is the fraction of experiments that reach an outcome A — if the experiment is repeated an infinite number of times:

$$P(A) = \lim_{n \rightarrow \infty} \frac{\text{Number of Results } A}{n}$$

- Benefits:

- “Objective” (not really... more on Lecture 8)
- Experimentally Verifiable

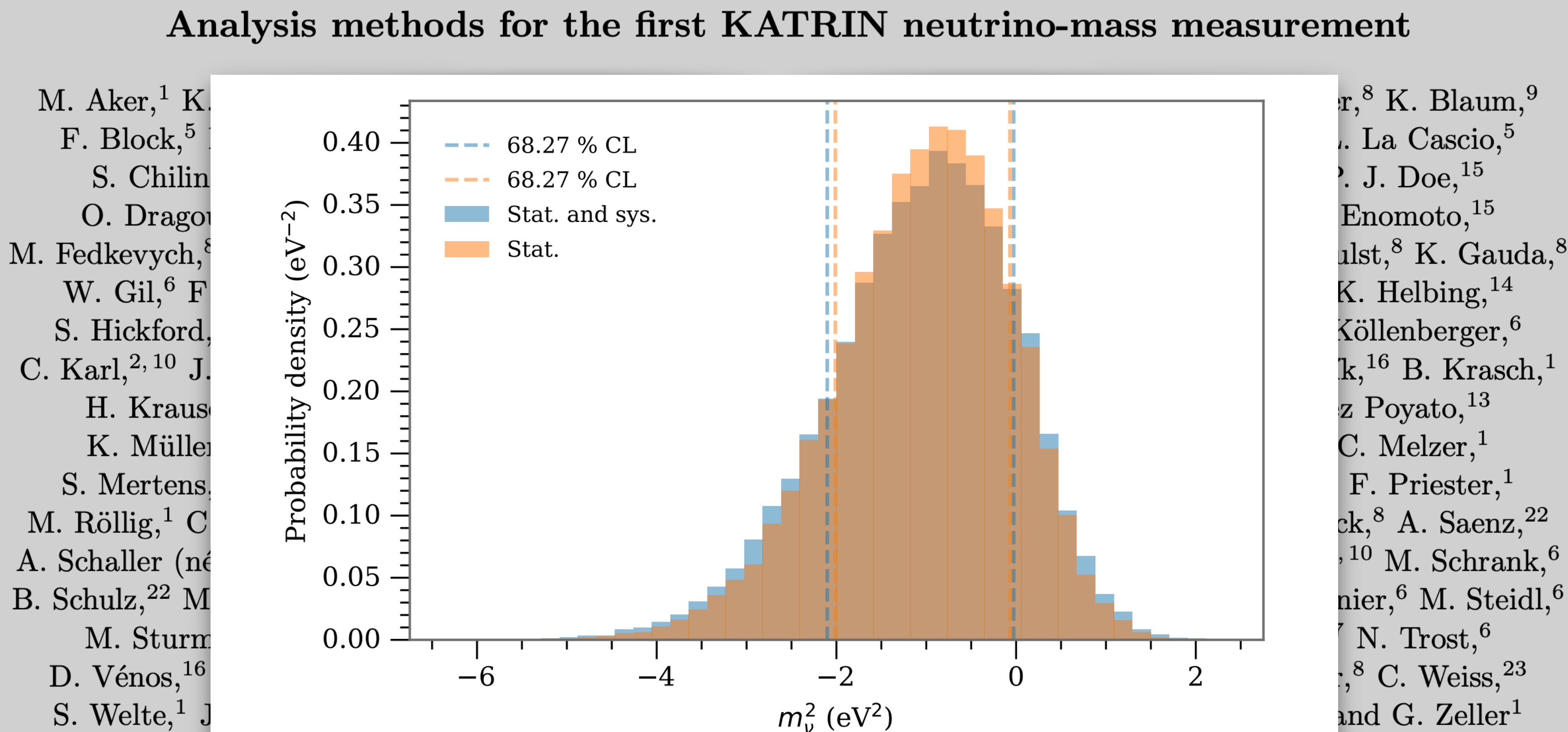
- Problems:

- Not all experiments are repeatable
- The question we want is “The Probability that A is true”

Interpretations of Statistics

Frequentist Statistics are sometimes counterintuitive

13 [hep-ex] 12 May 2021



Interpretations of Statistics

Bayesian Statistics

- The degree of belief in the hypothesis that A is true.
- Benefits:
 - It is the questions that we really want to answer
- Problems:
 - The apparent subjectivity of the prior makes some scientists reluctant to adopt it.

The diagram illustrates Bayes' Theorem with the following components and annotations:

- LIKELIHOOD** (orange text): the probability of "B" being TRUE given that "A" is TRUE. An arrow points from this text to the $P(B|A)$ term in the numerator.
- PRIOR** (teal text): the probability of "A" being TRUE. An arrow points from this text to the $P(A)$ term in the numerator.
- POSTERIOR** (green text): the probability of "A" being TRUE given that "B" is TRUE. An arrow points from this text to the $P(A|B)$ term in the denominator.
- The probability of "B" being TRUE** (pink text): An arrow points from this text to the $P(B)$ term in the denominator.

The equation is written as:

$$P(A|B) = \frac{P(B|A) P(A)}{P(B)}$$

Each term in the equation is enclosed in a colored box: $P(A|B)$ is green, $P(B|A)$ is orange, $P(A)$ is teal, and $P(B)$ is pink.

@luminousmen.com

Interpretations of Statistics

Bayesian Statistics is easier to interpret

[astro-ph.CO] 3 Apr 2025

Constraints on Neutrino Physics from DESI DR2 BAO and DR1 Full Shape

W. Elbers¹ A. Aviles^{2,3} H. E. Noriega^{2,4} D. Chebat⁵ A. Menegas¹ C. S. Frenk¹ C. Garcia-Quintero^{6,7}
D. Gonzalez⁸ M. S. Ahlen¹² O. Alves, A. Brodzeller¹⁹ D. Br F. J. Castander^{24,25} A. P. Cooper²⁷ A. A. Dey³⁰ B. Dey J. E. Forero-Romero Gontcho¹⁹ A. X. Gor C. Howlett⁴⁴ D. C. Lamman⁶ M. K. Lodha^{50,51} A. Meisner³⁰ J. Men E. Paillas⁵⁸ N. Pal F. Prada⁶⁵ I. Pérez H. Rivera-Morales⁴ J. E. Sanchez²⁹ D. S. G. Tarlé¹³ P. Tay B. A. Weaver,³⁰ A. M. Abdul-Karim⁵ Chi^{16,17} S. Brieden¹⁸ Rosell^{21,22} L. Casas,²³ Claybaugh,¹⁹ S. Cole¹ tia⁵ N. Deiosso²⁹ A. Font-Ribera²³ S. Gontcho A. Herrera-Alcantar^{43,5} r¹⁹ A. Kremin¹⁹ Levi¹⁹ Q. Li²⁸ Matthewson⁵⁰ J. A. Newman³² C. Poppett,^{19,63,64} skyi⁶ C. Ravoux⁶⁸ L. Samushia^{70,71,72} rayberry,³⁰ T. Tan⁵ M. Walther^{74,75} ou¹⁹ and H. Zou⁷⁷

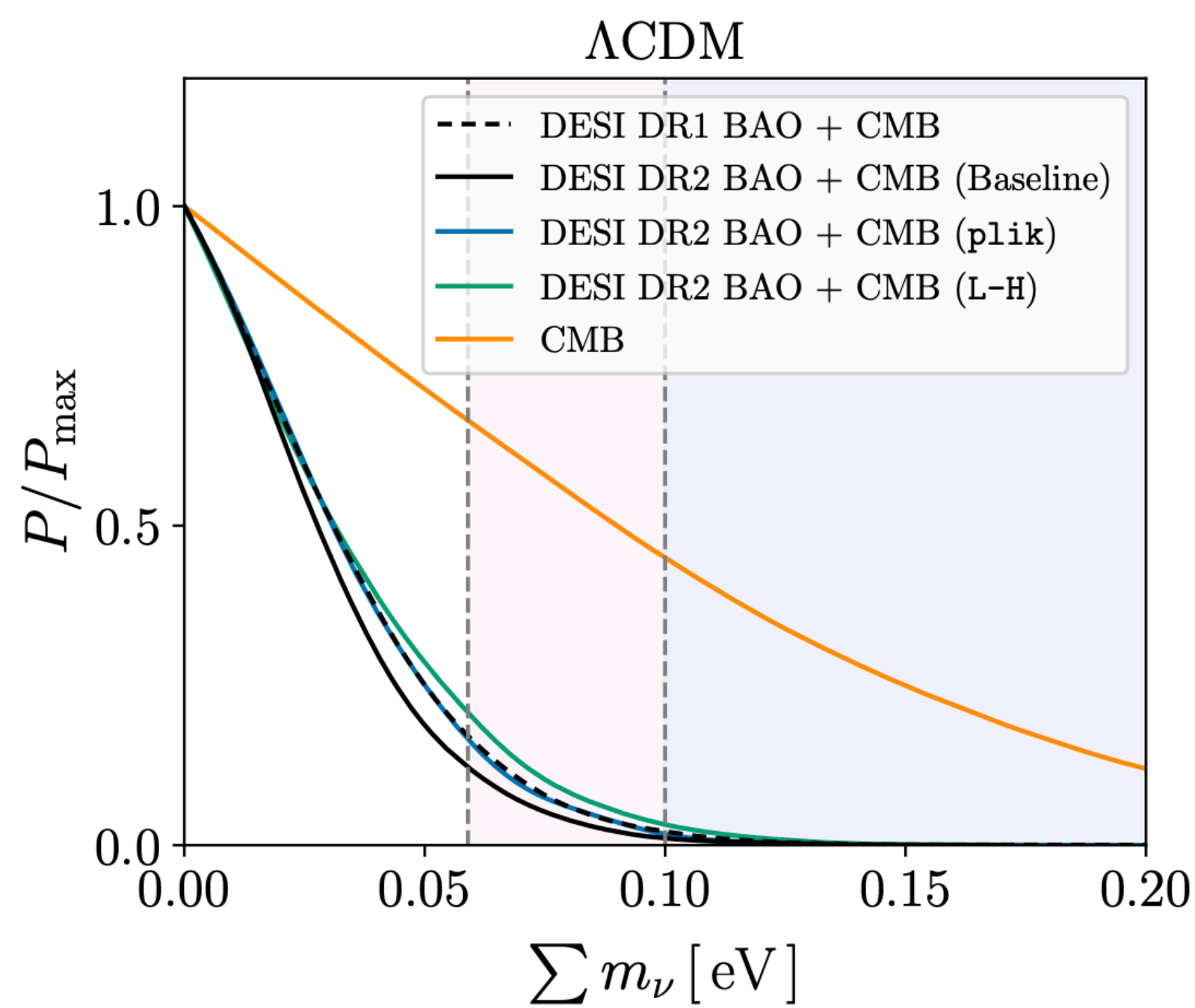
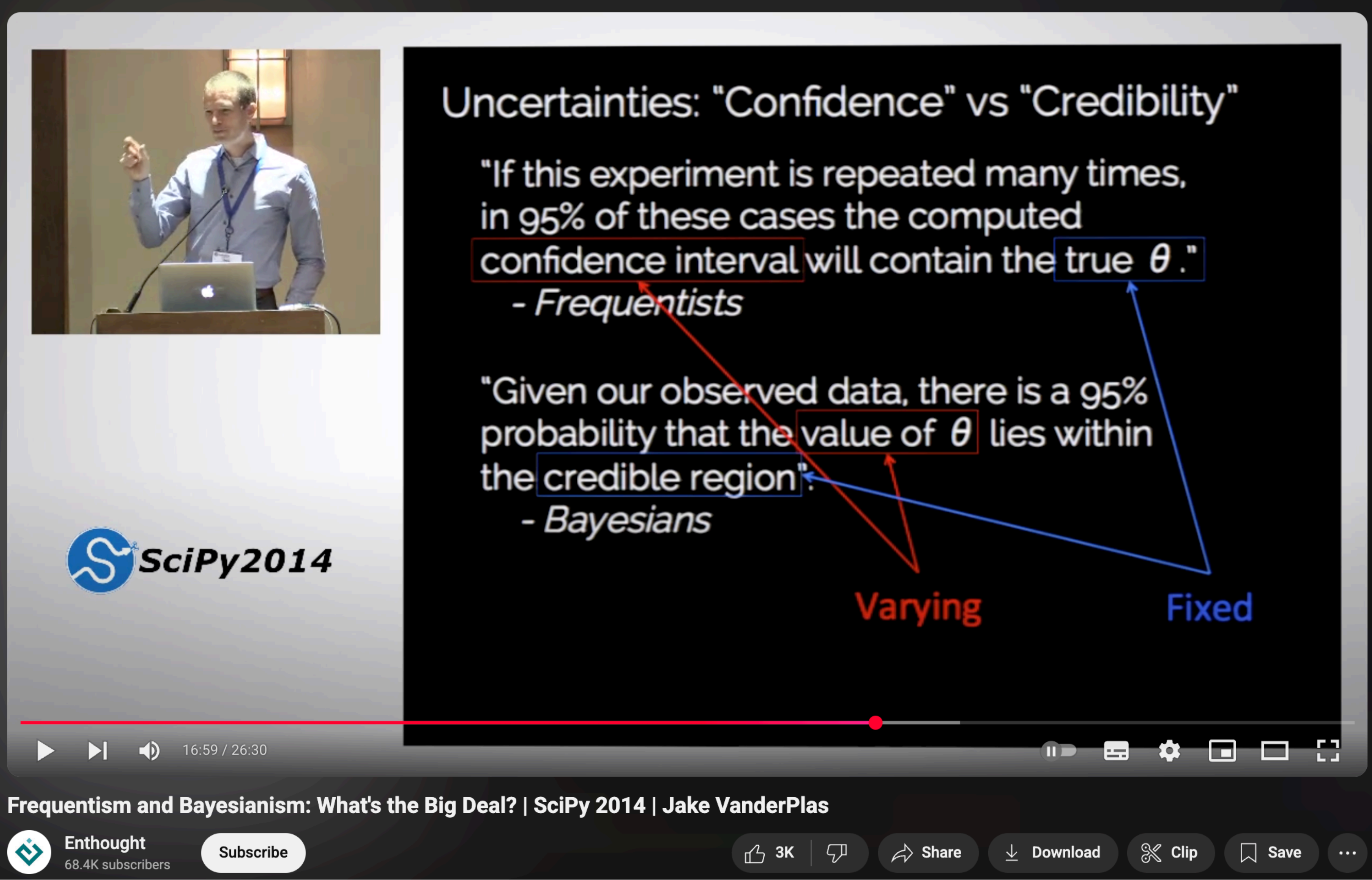


FIG. 4. Constraints on $\sum m_\nu$ from DESI DR2 BAO with different CMB likelihoods as well as for the baseline CMB dataset alone and combined with DR1 BAO. The vertical dashed lines and shaded regions indicate, from left to right, the minimum masses corresponding to the normal and inverted mass ordering scenarios.

Interpretations of Statistics



The video player shows a presentation slide titled "Uncertainties: 'Confidence' vs 'Credibility'". The slide compares two statistical interpretations:

- Frequentists:** "If this experiment is repeated many times, in 95% of these cases the computed confidence interval will contain the true θ ."
- Bayesians:** "Given our observed data, there is a 95% probability that the value of θ lies within the credible region."

Annotations on the slide include:

- A red arrow labeled "Varying" points from the "confidence interval" box to the "credible region" box.
- A blue arrow labeled "Fixed" points from the "true θ " box to the "value of θ " box.

The video player interface includes a SciPy 2014 logo, a progress bar at 16:59 / 26:30, and a title bar: "Frequentism and Bayesianism: What's the Big Deal? | SciPy 2014 | Jake VanderPlas". The channel is "Enthought" with 68.4K subscribers. Interaction buttons include 3K likes, a comment icon, Share, Download, Clip, Save, and a menu icon.

<https://youtu.be/KhAUfqhLakw>